

F5 Exponential equations

Name: _____

Class: _____

Date: _____

Time: **398 minutes**

Marks: **332 marks**

Comments:

EXAM PAPERS PRACTICE

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Q1.

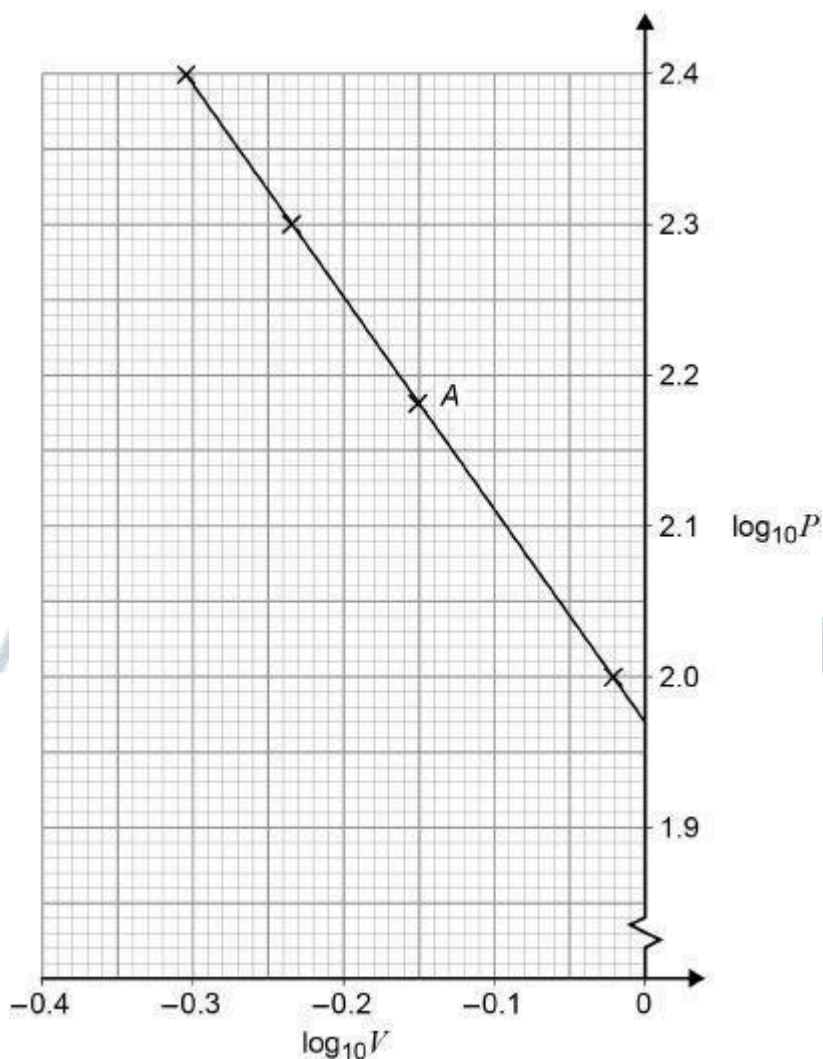
Maxine measures the pressure, P kilopascals, and the volume, V litres, in a fixed quantity of gas.

Maxine believes that the pressure and volume are connected by the equation

$$P = cV^d$$

where c and d are constants.

Using four experimental results, Maxine plots $\log_{10} P$ against $\log_{10} V$, as shown in the graph below.



- (a) Find the value of P and the value of V for the data point labelled A on the graph.

(2)

- (b) Calculate the value of each of the constants c and d .

(4)

- (c) Estimate the pressure of the gas when the volume is 2 litres.

(2)

(Total 8 marks)

Q2.

Trees in a forest may be affected by one of two types of fungal disease, but not by both.

The number of trees affected by disease A, n_A , can be modelled by the formula

$$n_A = ae^{0.1t}$$

where t is the time in years after 1 January 2017.

The number of trees affected by disease B, n_B , can be modelled by the formula

$$n_B = be^{0.2t}$$

On 1 January 2017 a **total** of 290 trees were affected by a fungal disease.

On 1 January 2018 a **total** of 331 trees were affected by a fungal disease.

- (a) Show that $b = 90$, to the nearest integer, and find the value of a . (3)
- (b) Estimate the total number of trees that will be affected by a fungal disease on 1 January 2020. (1)
- (c) Find the year in which the number of trees affected by disease B will first exceed the number affected by disease A. (3)
- (d) Comment on the long-term accuracy of the model. (1)
- (Total 8 marks)

Q3.

The function f is defined by $f(x) = e^{x-4}$, $x \in \mathbb{R}$

Find $f^{-1}(x)$ and state its domain.

(Total 3 marks)

Q4.

A curve is defined by the parametric equations

$$x = 4 \times 2^{-t} + 3$$

$$y = 3 \times 2^t - 5$$

- (a) Show that $\frac{dy}{dx} = -\frac{3}{4} \times 2^{2t}$ (3)
- (b) Find the Cartesian equation of the curve in the form $xy + ax + by = c$, where a , b and c are integers. (3)
- (Total 6 marks)

Q5.

A curve has equation $y = e^{2x}$

Find the coordinates of the point on the curve where the gradient of the curve is $\frac{1}{2}$
 Give your answer in an exact form.

(Total 5 marks)

Q6.

David has been investigating the population of rabbits on an island during a three-year period.

Based on data that he has collected, David decides to model the population of rabbits, R , by the formula

$$R = 50e^{0.5t}$$

where t is the time in years after 1 January 2016.

- (a) Using David's model:
- (i) state the population of rabbits on the island on 1 January 2016; (1)
 - (ii) predict the population of rabbits on 1 January 2021. (1)
- (b) Use David's model to find the value of t when $R = 150$, giving your answer to three significant figures. (2)
- (c) Give **one** reason why David's model may **not** be appropriate. (1)
- (d) On the same island, the population of crickets, C , can be modelled by the formula

$$C = 1000e^{0.1t}$$

where t is the time in years after 1 January 2016.

Using the two models, find the year during which the population of rabbits first exceeds the population of crickets.

(3)

(Total 8 marks)

Q7.

The function f is defined by

$$f(x) = 4 + 3^{-x}, x \in \mathbb{R}$$

- (a) Using set notation, state the range of f

(2)

- (b) The inverse of f is f^{-1}

- (i) Using set notation, state the domain of f^{-1}

(1)

- (ii) Find an expression for $f^{-1}(x)$

(3)

- (c) The function g is defined by

$$g(x) = 5 - \sqrt{x}, (x \in \mathbb{R} : x > 0)$$

- (i) Find an expression for $gf(x)$

(1)

- (ii) Solve the equation $gf(x) = 2$, giving your answer in an exact form.

(3)

(Total 10 marks)

Q8.

- (a) Three consecutive terms in an arithmetic sequence are $3e^{-p}$, 5 , $3e^p$

Find the possible values of p . Give your answers in an exact form.

(6)

- (b) Prove that there is no possible value of q for which $3e^{-q}$, 5 , $3e^q$ are consecutive terms of a geometric sequence.

(4)

(Total 10 marks)

Q9.

At time $t = 0$, a parachutist jumps out of an airplane that is travelling horizontally.

The velocity, v m s⁻¹, of the parachutist at time t seconds is given by:

$$\mathbf{v} = (40e^{-0.2t})\mathbf{i} + 50(e^{-0.2t} - 1)\mathbf{j}$$

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that the parachutist is at the origin when $t = 0$

Model the parachutist as a particle.

- (a) Find an expression for the position vector of the parachutist at time t .

(4)

- (b) The parachutist opens her parachute when she has travelled 100 metres horizontally.

Find the vertical displacement of the parachutist from the origin when she opens her parachute.

(4)

- (c) Carefully, explaining the steps that you take, deduce the value of g used in the formulation of this model.

(3)

(Total 11 marks)

Q10.

A circular ornamental garden pond, of radius 2 metres, has weed starting to grow and cover its surface.

As the weed grows, it covers an area of A square metres. A simple model assumes that the weed grows so that the rate of increase of its area is proportional to A .

- (a) Show that the area covered by the weed can be modelled by

$$A = Be^{kt}$$

where B and k are constants and t is time in days since the weed was first noticed.

(4)

- (b) When it was first noticed, the weed covered an area of 0.25 m^2 . Twenty days later the weed covered an area of 0.5 m^2

- (i) State the value of B .

(1)

- (ii) Show that the the model for the area covered by the weed can be written as

$$A = 2^{\frac{t}{20} - 2}$$

(4)

- (iii) How many days does it take for the weed to cover half of the surface of the pond?

(2)

- (c) State one limitation of the model.

(1)

- (d) Suggest one refinement that could be made to improve the model.

(1)

(Total 13 marks)

Q11.

- (a) Describe a geometrical transformation that maps the graph of $y = 4^x$ onto the graph of $y = 3 \times 4^x$.

(2)

- (b) Sketch the curve with equation $y = 3 \times 4^x$, indicating the value of the intercept on the y -axis.

(2)

- (a) The curve with equation $y = 4^{-x}$ intersects the curve $y = 3 \times 4^x$ at the point P . Use logarithms to find the x -coordinate of P , giving your answer to three significant figures.

(5)

(Total 9 marks)

Q12.

A biologist is investigating the growth of a population of a species of rodent. The biologist proposes the model

$$N = \frac{500}{1 + 9e^{-\frac{t}{8}}}$$

for the number of rodents, N , in the population t weeks after the start of the investigation.

Use this model to answer the following questions.

- (a) (i) Find the size of the population at the start of the investigation.

(1)

- (ii) Find the size of the population 24 weeks after the start of the investigation. Give your answer to the nearest whole number.

(1)

- (iii) Find the number of weeks that it will take the population to reach 400. Give your answer in the form $t = r \ln s$, where r and s are integers.

(3)

- (b) (i) Show that the rate of growth, $\frac{dN}{dt}$, is given by

$$\frac{dN}{dt} = \frac{N}{4000} (500 - N)$$

(4)

- (ii) The maximum rate of growth occurs after T weeks. Find the value of T .

(4)

(Total 13 marks)

Q13.

- (a) Sketch the graph of $y = 9^x$, indicating the value of the intercept on the y -axis. (2)
- (b) Use logarithms to solve the equation $9^x = 15$, giving your value of x to three significant figures. (2)
- (c) The curve $y = 9^x$ is reflected in the y -axis to give the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (1)

(Total 5 marks)

Q14.

- (a) Sketch the graph of $y = \frac{1}{2^x}$, indicating the value of the intercept on the y -axis. (2)
- (b) Use logarithms to solve the equation $\frac{1}{2^x} = \frac{5}{4}$, giving your answer to three significant figures. (3)
- (c) Given that

$$\log_a(b^2) + 3 \log_a y = 3 + 2 \log_a \left(\frac{y}{a} \right)$$

express y in terms of a and b .

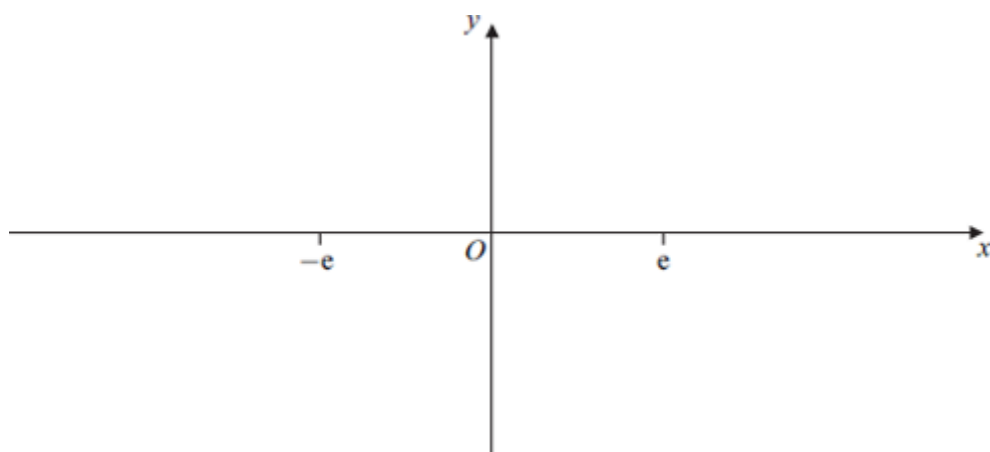
Give your answer in a form not involving logarithms.

(5)

(Total 10 marks)

Q15.

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = \ln x$ onto the graph of $y = 4 \ln(x - e)$. (4)
- (b) Sketch, on the axes given below, the graph of $y = |4 \ln(x - e)|$, indicating the exact value of the x -coordinate where the curve meets the x -axis.



(3)

- (c) (i) Solve the equation $|4 \ln(x - e)| = 4$.

(3)

- (ii) Hence, or otherwise, solve the inequality $|4 \ln(x - e)| \geq 4$.

(3)

(Total 13 marks)

Q16.

A scientist is testing models for the growth and decay of colonies of bacteria.

For a particular colony, which is growing, the model is $P = Ae^{\frac{1}{8}t}$, where P is the number of bacteria after a time t minutes and A is a constant.

- (a) This growing colony consists initially of 500 bacteria. Calculate the number of bacteria, according to the model, after one hour. Give your answer to the nearest thousand.

(2)

- (b) For a second colony, which is decaying, the model is $Q = 500\,000e^{-\frac{1}{8}t}$, where Q is the number of bacteria after a time t minutes.

Initially, the growing colony has 500 bacteria and, at the same time, the decaying colony has 500 000 bacteria.

- (i) Find the time at which the populations of the two colonies will be equal, giving your answer to the nearest 0.1 of a minute.

(3)

- (ii) The population of the growing colony will exceed that of the decaying colony by 45 000 bacteria at time T minutes.

Show that

$$\left(e^{\frac{1}{8}T}\right)^2 - 90e^{\frac{1}{8}T} - 1000 = 0$$

and hence find the value of T , giving your answer to one decimal place.

(4)

(Total 9 marks)

Q17.

- (a) Sketch the curve with equation $y = 7^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) The curve C_1 has equation $y = 7^x$.
The curve C_2 has equation $y = 7^{2x} - 12$.

- (i) By forming and solving a quadratic equation, prove that the curves C_1 and C_2 intersect at exactly one point. State the y -coordinate of this point.

(4)

- (ii) Use logarithms to find the x -coordinate of the point of intersection of C_1 and C_2 , giving your answer to three significant figures.

(2)

(Total 8 marks)

Q18.

The value, $\pounds V$, of an initial investment, $\pounds P$, at the end of n years is given by the formula

$$V = P \left(1 + \frac{r}{100}\right)^n$$

where $r\%$ per year is the fixed interest rate.

Mr Brown invests $\pounds 1000$ in Barcelona Bank at a fixed interest rate of 3% per year.

- (a) (i) Find the value of Mr Brown's investment at the end of 5 years. Give your value to the nearest $\pounds 10$.

(1)

- (ii) The value of Mr Brown's investment will first exceed $\pounds 2000$ after N complete years.

Find the value of N .

(3)

- (b) Mrs White invests $\pounds 1500$ in Bilbao Bank at a fixed interest rate of 1.5% per year. Mr Brown and Mrs White invest their money at the same time. The value of Mr Brown's investment will first exceed the value of Mrs White's investment after T complete years.

Find the value of T .

(4)

(Total 8 marks)

Q19.

A geometric series has third term 36 and sixth term 972.

- (a) (i) Show that the common ratio of the series is 3.

(2)

- (ii) Find the first term of the series.

(2)

- (b) The n th term of the series is u_n .

- (i) Show that $\sum_{n=1}^{20} u_n = 2(3^{20} - 1)$.

(2)

- (ii) Find the least value of n such that $u_n > 4 \times 10^{15}$.

(3)

(Total 9 marks)

Q20.

A model for the radioactive decay of a form of iodine is given by

$$m = m_0 2^{-\frac{1}{8}t}$$

The mass of the iodine after t days is m grams. Its initial mass is m_0 grams.

- (a) Use the given model to find the mass that remains after 10 grams of this form of iodine have decayed for 14 days, giving your answer to the nearest gram.

(2)

- (b) A mass of m_0 grams of this form of iodine decays to $\frac{m_0}{16}$ grams in d days.

Find the value of d .

(2)

- (c) After n days, a mass of this form of iodine has decayed to less than 1% of its initial mass.

Find the minimum integer value of n .

(3)

(Total 7 marks)

Q21.

- (a) Sketch the curve with equation $y = 4^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) Describe the geometrical transformation that maps the graph of $y = 4^x$ onto the graph of $y = 4^x - 5$.

(2)

- (c) (i) Use the substitution $Y = 2^x$ to show that the equation $4^x - 2^{x+2} - 5 = 0$ can be written as $Y^2 - 4Y - 5 = 0$.

(2)

- (ii) Hence show that the equation $4^x - 2^{x+2} - 5 = 0$ has only one real solution.

Use logarithms to find this solution, giving your answer to three decimal places.

(4)

(Total 10 marks)

Q22.

The average weekly pay of a footballer at a certain club was £80 on 1 August 1960. By 1 August 1985, this had risen to £2000.

The average weekly pay of a footballer at this club can be modelled by the equation

$$P = Ak^t$$

where £ P is the average weekly pay t years after 1 August 1960, and A and k are constants.

- (a) (i) Write down the value of A .

(1)

- (ii) Show that the value of k is 1.137411, correct to six decimal places.

(2)

- (b) Use this model to predict the year in which, on 1 August, the average weekly pay of a footballer at this club will first exceed £100 000.

(3)

(Total 6 marks)

Q23.

- (a) Sketch the curve with equation $y = 2^x$, indicating the coordinates of any point where the curve intersects the coordinate axes.

(2)

- (b) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate

value for $\int_0^2 2^x \, dx$, giving your answer to three significant figures.

(4)

- (ii) State how you could obtain a better approximation to the value of the integral using the trapezium rule.

(1)

- (c) Describe a geometrical transformation that maps the graph of $y = 2^x$ onto the graph of $y = 2^{x+7} + 3$.

(3)

- (d) The curve $y = 2^{x+k} + 3$ intersects the y -axis at the point $A(0, 8)$.

Show that $k = \log_m n$, where m and n are integers.

(2)

(Total 12 marks)

Q24.

A botanist is investigating the rate of growth of a certain species of toadstool. She observes that a particular toadstool of this type has a height of 57 millimetres at a time 12 hours after it begins to grow.

$$h = A \left(1 - e^{-\frac{1}{4}t} \right)$$

She proposes the model, where A is a constant, for the height h millimetres of the toadstool, t hours after it begins to grow.

- (a) Use this model to:

- (i) find the height of the toadstool when $t = 0$;

(1)

- (ii) show that $A = 60$, correct to two significant figures.

(2)

$$h = 60 \left(1 - e^{-\frac{1}{4}t} \right)$$

- (b) Use the model to:

- (i) show that the time T hours for the toadstool to grow to a height of 48 millimetres is given by

$$T = a \ln b$$

where a and b are integers;

(3)

(ii) show that $\frac{dh}{dt} = 15 - \frac{h}{4}$;

(3)

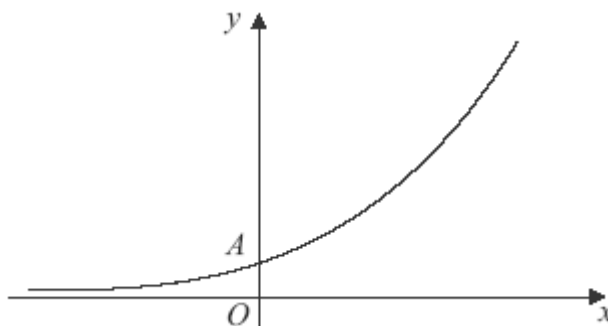
- (iii) find the height of the toadstool when it is growing at a rate of 13 millimetres per hour.

(1)

(Total 10 marks)

Q25.

The diagram shows a sketch of the curve $y = 2^{4x}$.



The curve intersects the y -axis at the point A .

- (a) Find the value of the y -coordinate of A .

(1)

- (b) Use the trapezium rule with six ordinates (five strips) to find an approximate value for $\int_0^1 2^{4x} dx$, giving your answer to two decimal places.

(4)

- (c) Describe the geometrical transformation that maps the graph of $y = 2^{4x}$ onto the graph of $y = 2^{4x-3}$.

(2)

- (d) The curve $y = 2^{4x}$ is translated by the vector $\begin{bmatrix} 1 \\ -\frac{1}{2} \end{bmatrix}$ to give the curve $y = g(x)$.

The curve $y = g(x)$ crosses the x -axis at the point Q . Find the x -coordinate of Q .

(4)

- (e) (i) Given that

$$\log_a k = 3 \log_a 2 + \log_a 5 - \log_a 4$$

show that $k = 10$.

(3)

- (ii) The line $y = \frac{5}{4}$ crosses the curve $y = 2^{4x-3}$ at the point P . Show that the

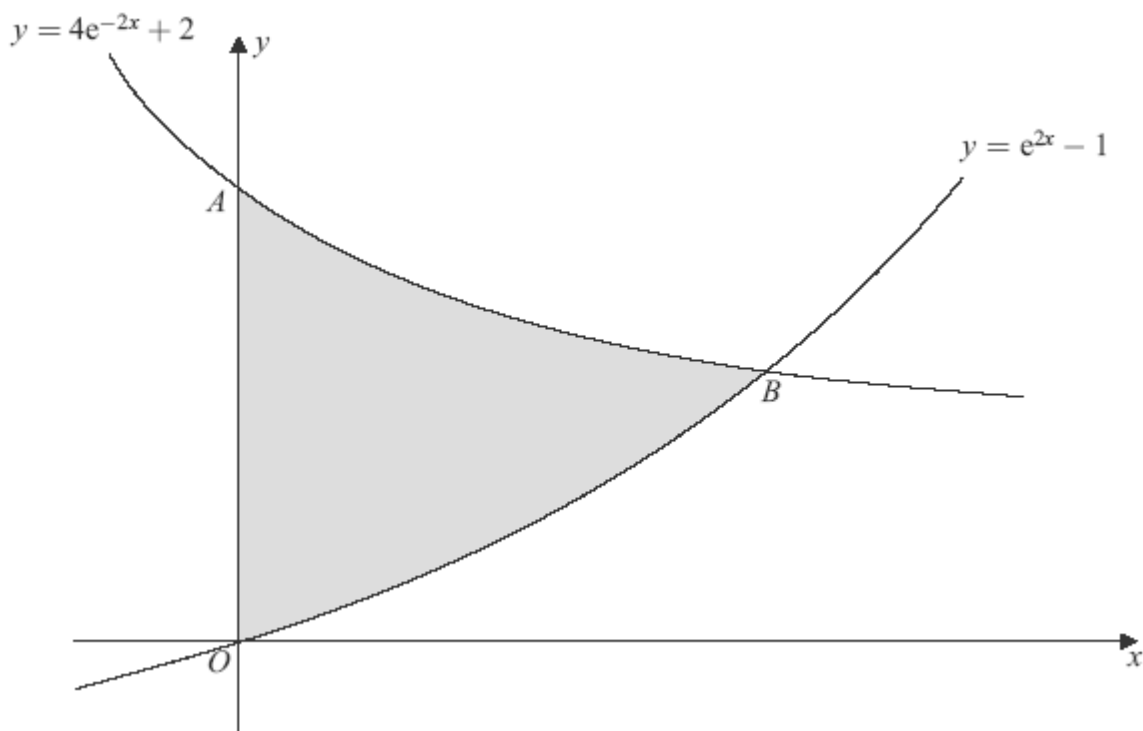
$$x\text{-coordinate of } P \text{ is } \frac{1}{4 \log_{10} 2}.$$

(3)

(Total 17 marks)

Q26.

The diagram shows the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$.



The curve $y = 4e^{-2x} + 2$ crosses the y -axis at the point A and the curves intersect at the point B .

- (a) Describe a sequence of two geometrical transformations that maps the graph of $y = e^x$ onto the graph of $y = e^{2x} - 1$.

(4)

- (b) Write down the coordinates of the point A .

(1)

- (c) (i) Show that the x -coordinate of the point B satisfies the equation

$$(e^{2x})^2 - 3e^{2x} - 4 = 0$$

(2)

(ii) Hence find the exact value of the x -coordinate of the point B .

(3)

- (d) Find the exact value of the area of the shaded region bounded by the curves $y = e^{2x} - 1$ and $y = 4e^{-2x} + 2$ and the y -axis.

(5)

(Total 15 marks)

Q27.

- (a) Write each of the following in the form $\log_a k$, where k is an integer:

(i) $\log_a 4 + \log_a 10$;

(1)

(ii) $\log_a 16 - \log_a 2$;

(1)

(iii) $3 \log_a 5$.

(1)

- (b) Use logarithms to solve the equation $(1.5)^{3x} = 7.5$, giving your value of x to three decimal places.

(3)

- (c) Given that $\log_2 p = m$ and $\log_8 q = n$, express pq in the form 2^y , where y is an expression in m and n .

(3)

(Total 9 marks)

Q28.

- (a) Given that $3e^x = 4$, find the exact value of x .

(2)

- (b) (i) By substituting $y = e^x$, show that the equation $3e^x + 20e^{-x} = 19$ can be written as $3y^2 - 19y + 20 = 0$.

(1)

- (ii) Hence solve the equation $3e^x + 20e^{-x} = 19$, giving your answers as exact values.

(3)

(Total 6 marks)

Q29.

- (a) (i) Find the value of p for which $\sqrt{125} = 5^p$.

(2)

(ii) Hence solve the equation $5^{2x} = \sqrt{125}$.

(1)

(b) Use logarithms to solve the equation $3^{2x-1} = 0.05$, giving your value of x to four decimal places.

(3)

(c) It is given that

$$\log_a x = 2(\log_a 3 + \log_a 2) - 1$$

Express x in terms of a , giving your answer in a form not involving logarithms.

(4)

(Total 10 marks)

Q30.

A car depreciates in value according to the model

$$V = Ak^t$$

where $\pounds V$ is the value of the car t months from when it was new, and A and k are constants.

Its value when new was $\pounds 12\,499$ and 36 months later its value was $\pounds 7000$.

(a) (i) Write down the value of A .

(1)

(ii) Show that the value of k is 0.984 025, correct to six decimal places.

(2)

(b) The value of this car first dropped below $\pounds 5000$ during the n th month from new. Find the value of n .

(3)

(Total 6 marks)

Q31.

(a) Sketch the graph of $y = 3^x$, stating the coordinates of the point where the graph crosses the y -axis.

(2)

(b) Describe a single geometrical transformation that maps the graph of $y = 3^x$:

(i) onto the graph of $y = 3^{2x}$;

(2)

(ii) onto the graph of $y = 3^{x+1}$.

(2)

- (c) (i) Using the substitution $Y = 3^x$, show that the equation

$$9^x - 3^{x+1} + 2 = 0$$

can be written as

$$(Y - 1)(Y - 2) = 0$$

(2)

- (ii) Hence show that the equation $9^x - 3^{x+1} + 2 = 0$ has a solution $x = 0$ and, by using logarithms, find the other solution, giving your answer to four decimal places.

(4)

(Total 12 marks)

Q32.

David is researching changes in the selling price of houses. One particular house was sold on 1 January 1885 for £20. Sixty years later, on 1 January 1945, it was sold for £2000. David proposes a model

$$P = Ak^t$$

for the selling price, £ P , of this house, where t is the time in years after 1 January 1885 and A and k are constants.

- (a) (i) Write down the value of A .

(1)

- (ii) Show that, to six decimal places, $k = 1.079775$.

(2)

- (iii) Use the model, with this value of k , to estimate the selling price of this house on 1 January 2008. Give your answer to the nearest £1000.

(2)

- (b) For another house, which was sold for £15 on 1 January 1885, David proposes the model

$$Q = 15 \times 1.082709^t$$

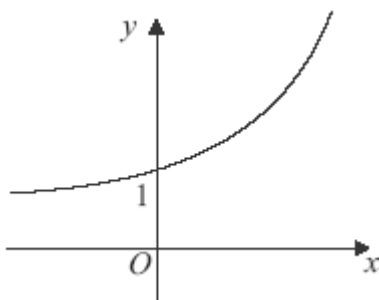
for the selling price, £ Q , of this house t years after 1 January 1885. Calculate the year in which, according to these models, these two houses would have had the same selling price.

(4)

(Total 9 marks)

Q33.

The diagram shows a sketch of the curve with equation $y = 6^x$.



- (a) (i) Use the trapezium rule with five ordinates (four strips) to find an approximate value for $\int_0^2 6^x dx$, giving your answer to three significant figures. (4)
- (ii) Explain, with the aid of a diagram, whether your approximate value will be an overestimate or an underestimate of the true value of $\int_0^2 6^x dx$. (2)
- (b) (i) Describe a single geometrical transformation that maps the graph of $y = 6^x$ onto the graph of $y = 6^{3x}$. (2)
- (ii) The line $y = 84$ intersects the curve $y = 6^{3x}$ at the point A . By using logarithms, find the x -coordinate of A , giving your answer to three decimal places. (4)
- (c) The graph of $y = 6^x$ is translated by $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$ to give the graph of the curve with equation $y = f(x)$. Write down an expression for $f(x)$. (2)

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(Total 14 marks)

Q34.

- (a) The number of fish in a lake is decreasing. After t years, there are x fish in the lake. The rate of decrease of the number of fish is proportional to the number of fish currently in the lake.
- (i) Formulate a differential equation, in the variables x and t and a constant of proportionality k , where $k > 0$, to model the rate at which the number of fish in the lake is decreasing. (2)
- (ii) At a certain time, there were 20 000 fish in the lake and the rate of decrease was 500 fish per year. Find the value of k . (2)

- (b) The equation

$$P = 2000 - Ae^{-0.05t}$$

is proposed as a model for the number of fish, P , in another lake, where t is the time in years and A is a positive constant.

On 1 January 2008, a biologist estimated that there were 700 fish in this lake.

- (i) Taking 1 January 2008 as $t = 0$, find the value of A .

(1)

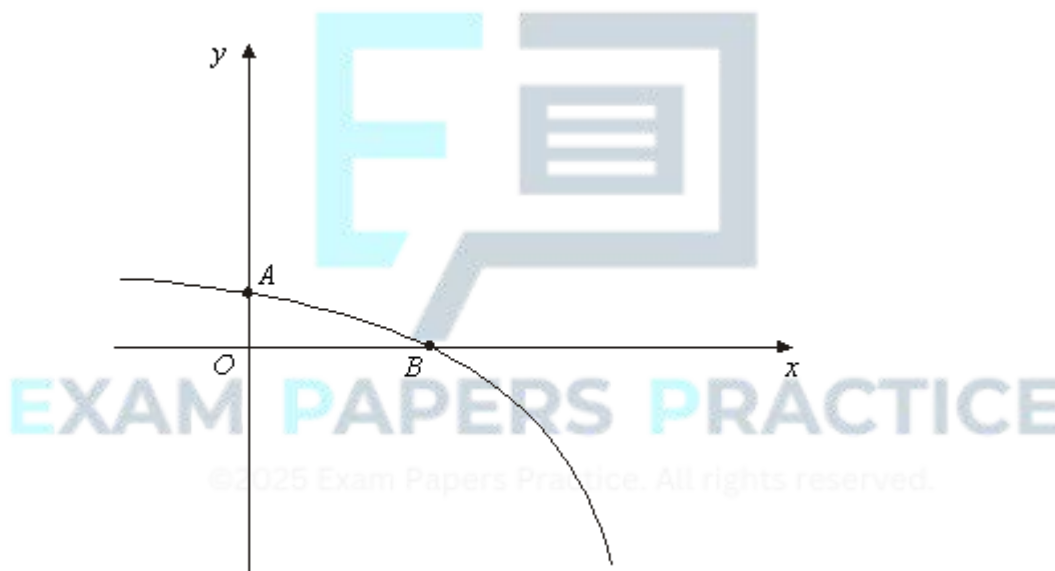
- (ii) Hence find the year during which, according to this model, the number of fish in this lake will first exceed 1900.

(4)

(Total 9 marks)

Q35.

The sketch shows the graph of $y = 4 - e^{2x}$. The curve crosses the y -axis at the point A and the x -axis at the point B .



- (a) (i) Find $\int (4 - e^{2x}) dx$.

(2)

- (ii) Hence show that $\int_0^{\ln 2} (4 - e^{2x}) dx = 4 \ln 2 - \frac{3}{2}$.

(2)

- (b) (i) Write down the y -coordinate of A .

(1)

- (ii) Show that $x = \ln 2$ at B .

(2)

- (c) Find the equation of the normal to the curve $y = 4 - e^{2x}$ at the point B .

(4)

- (d) Find the area of the region enclosed by the curve $y = 4 - e^{2x}$, the normal to the curve at B and the y -axis.

(3)

(Total 14 marks)



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